

Software Testing Intern Assignment

TEST PLAN: NOVA - TEAM PRODUCTIVITY PLATFORM

1. INTRODUCTION

1.1 Document Purpose

This document outlines the testing strategy, objectives, and scope for the production release of NOVA, a project management web application.

1.2 Project Overview

NOVA is a project management tool where users can manage profiles, register/login, create projects, assign tasks, view dashboards, and receive notifications.

2. TEST OBJECTIVES

- Verify all core application modules function as per requirements.
- Ensure early identification of functional, UI, usability, API, and regression issues.
- Deliver clean execution and test artifacts to recommend or block production release.

3. SCOPE OF TESTING

3.1 In-Scope (Modules to Test)

- Authentication: Registration, Login, Logout, Forgot Password, Password Reset.
- User Management: Profile, Update profile, Change password, User roles.
- Project Management: Create, Edit, Delete, View projects, Add members.
- Task Management: Create, Edit, Delete, Assign tasks, Change priority/status, Set due dates, Comments.
- Dashboard: Project/Task metrics and counters.
- Search & Filters: Project/Task search and multi-attribute filtering.
- Notifications: Alerts for assignments, completions, due dates, and updates.

3.2 Out-of-Scope

- Performance/Load testing under high concurrent volume (unless localized Lighthouse/DevTools auditing is required).
- Security penetration testing.

4. TEST APPROACH & METHODOLOGY

- Manual Testing: Exploratory and functional end-to-end testing to discover hidden UI and usability bugs.
- API Testing: Component validation using tools like Postman/Insomnia.
- Automation Testing: UI automation for the core Smoke and Regression suites using Java, Selenium, and TestNG.

5. TEST ENVIRONMENT & TOOLS

- Test Management: Google Sheets / Excel
- Automation Framework: Java + Selenium WebDriver + TestNG
- API Testing: Postman
- Browsers: Google Chrome, Mozilla Firefox, Microsoft Edge

6. CRITERIA

6.1 Entry Criteria

- QA assignment requirements and mock/live environment details are available.
- Initial manual test cases and automated scripts are drafted.

6.2 Exit Criteria

- Minimum 50 test scenarios and 30 detailed test cases are executed.
- At least 5 API tests and 5 automated scripts run successfully.
- All critical and major bugs are reported, retested, and resolved (or documented for release recommendation).

7. DELIVERABLES

- Test Plan Document
- Test Case Sheet & Requirement Traceability Matrix (RTM)
- Bug Reports & Test Evidence
- TestNG Automation Code Suite
- Final QA Report

TEST SCENARIO:

Module 1: Authentication Test Scenarios

User Registration (Sign Up)

- **TS_01:** Verify registration with valid details (First Name, Last Name, Email, strong Password).
- **TS_02:** Verify registration fails with an already registered email address.
- **TS_03:** Verify validation errors when mandatory fields are left blank.
- **TS_04:** Verify password strength validation (e.g., rejecting weak passwords lacking special characters/numbers).
- **TS_05:** Verify behavior when "Password" and "Confirm Password" fields do not match.
- **TS_06:** Verify field-level validation for invalid email formats (e.g., missing @ or domain).

User Login

- **TS_07:** Verify successful login using valid credentials.
- **TS_08:** Verify login fails with an incorrect password.
- **TS_09:** Verify login fails with an unregistered email address.
- **TS_10:** Verify validation messaging when the login form is submitted empty.
- **TS_11:** Verify the functionality of the "Remember Me" checkbox on subsequent sessions.
- **TS_12:** Verify account lockout behavior after multiple consecutive failed login attempts.

Forgot / Reset Password

- **TS_13:** Verify a password reset link is successfully sent to a registered email.
- **TS_14:** Verify error message when requesting a reset link for an unregistered email.
- **TS_15:** Verify that the password reset token/link expires after its designated time limit.
- **TS_16:** Verify successful password update using the valid reset link.

User Logout

- **TS_17:** Verify clicking "Logout" successfully terminates the user session.
- **TS_18:** Verify that navigating back via the browser after logging out does not expose authenticated pages.

Module 2: User Management

- **TS_19:** Verify that a user can update their profile information (First Name, Last Name, Profile Picture).
- **TS_20:** Verify that updating the profile with invalid data formats triggers proper error messages.
- **TS_21:** Verify that a user can successfully change their password by providing the correct current password.
- **TS_22:** Verify that a password change fails if the current password is input incorrectly.
- **TS_23:** Verify that permissions shift correctly when a user's role is updated (e.g., Admin vs. Regular User).
- **TS_24:** Verify that restricted application modules are completely inaccessible to unauthorized user roles.

Module 3: Project Management

- **TS_25:** Verify that an authorized user can successfully create a new project with a unique name.
- **TS_26:** Verify that a project cannot be created with duplicate names or empty mandatory fields.
- **TS_27:** Verify that an existing project's details (Name, Description) can be edited and saved.
- **TS_28:** Verify that a project can be deleted, and a confirmation pop-up is displayed before deletion.
- **TS_29:** Verify that deleting a project cascades and clears out all associated tasks within it.
- **TS_30:** Verify that multiple team members can be successfully added to a project workspace.
- **TS_31:** Verify that a user can view the complete project overview and its team activity timeline.

Module 4: Task Management

- **TS_32:** Verify that a task can be successfully created inside a project with a title and description.
- **TS_33:** Verify that a task can be assigned to an active project team member.
- **TS_34:** Verify changing a task's priority level (Low, Medium, High) updates the task view instantly.
- **TS_35:** Verify changing a task's status (To Do, In Progress, Review, Done) updates across boards.
- **TS_36:** Verify that setting a valid task due date locks correctly and triggers appropriate tags.
- **TS_37:** Verify that setting a past date as a due date handles validation correctly or flags it as overdue.
- **TS_38:** Verify that a task can be successfully deleted, removing it from all active views.
- **TS_39:** Verify that team members can post and render text comments on a specific task thread.

Module 5: Dashboard

- **TS_40:** Verify that the "Total Projects" and "Active Projects" counters update accurately upon project changes.
- **TS_41:** Verify that the "Open Tasks" and "Completed Tasks" metrics accurately track task state changes.

- **TS_42:** Verify that the "Overdue Tasks" metric dynamically includes tasks past their due dates.
- **TS_43:** Verify clicking metric counters on the dashboard filters and redirects you to the matching item list.
- **TS_44:** Verify that dashboard layout renders uniformly without overlapping data elements across screen sizes.

Module 6: Search & Filters

- **TS_45:** Verify that searching a project or task by partial or full title yields accurate results.
- **TS_46:** Verify that filtering items by status, priority, or assignee strictly displays matching data.
- **TS_47:** Verify that applying multiple filters concurrently works as expected (e.g., High Priority + Assigned to Me).
- **TS_48:** Verify that searching for non-existent keywords returns an explicit "No Results Found" message.

Module 7: Notifications

- **TS_49:** Verify that an instant notification is triggered when a user is assigned a task or a project updates.
- **TS_50:** Verify that notifications for upcoming due dates trigger properly relative to the system clock.